

The problem of consciousness is less hard than you think

Abstract: What is consciousness? In this paper, we advance a thermodynamic-informational account of consciousness, grounded in the dynamics of systems maintained far from equilibrium. We propose that (a) the intensity of subjective experience scales with thermodynamic distance from equilibrium (operationalized via the normalized ratio $\mathcal{R}$) and (b) the qualitative character of this experience depends on the balance between informational integration Φ and entropy H . An Arrhenius-style corollary, in turn, links these quantities to a temperature-dependent metabolic flux (tracked by CMRO₂). Furthermore, $\mathcal{R}$ is modelled as a beta-distributed variable whose parameters encode the Φ – H balance, yielding a continuous space of conscious states that span highly integrated, low-entropy states (flow and meditative states) and disordered, high-entropy states (delirium and psychedelia). Within our account, conscious states are identified with regions of a joint thermodynamic-informational space in which Friston-style variational free energy F is minimized. Philosophically, our account reframes the hard problem of consciousness. Instead of positing or conceding an explanatory gap between biophysical processes and subjective experience, we treat consciousness as a characteristic regime of dissipative organization. Our central explanatory task therefore becomes one of regime specification: determining the boundary

conditions under which certain dissipative systems exhibit the structured, integrated, and flexible dynamics constitutive of subjective experience and an inner life.

Keywords: Consciousness; hard problem; thermodynamic; metabolism; Beta distribution; free energy

§1 Introduction

Contemporary theories of consciousness remain divided between two camps. On the one hand, structural, functional, or computational accounts seek to explain consciousness in terms of their biophysical organization or hardware (structure), causal roles and relations (function), or patterns of information processing (computation) (Albantakis et al., 2023; Block, 1995; Churchland, 2013; Crick, Koch, et al., 1990; Putnam, 1967; Searle, 2017; Tononi, Boly, et al., 2016; Turing, 1950). On the other hand, Chalmers’ (1995, 1996) formulation of the hard problem insists that no account of structure, function, or information-processing, however complete, will suffice to explain why these processes – biophysical, functional, or computational – are accompanied by subjective experience and an inner life at all.

In this paper, we attempt to dissolve the dichotomy between these two camps by nesting consciousness within a unified thermodynamic-informational description of living systems. In particular, we argue that subjective experience is not an ontologically independent addition to life-relevant processes, but rather a manifestation of certain dissipative structures that can maintain local order far from thermodynamic equilibrium. According to our view:

The intensity of conscious experience $C_{\text{intensity}}$ scales with ther-

modynamic distance $\mathcal{D}$ from equilibrium (§ 5.1);
 Conscious states are mediated in part by temperature-dependent
 changes in reaction kinetics and cerebral metabolism (§ 5.2);
 The quality and coherence of conscious experience depends on a
 balance between informational integration Φ and informational
 entropy H (§ 5.3);
 Conscious states can be modelled as a Beta-distributed variable
 over the Φ – H state space, reflecting the constrained trade-off be-
 tween integration Φ and entropy H (§ 5.4);
 Conscious states occupy regions in the Φ – H state space in which
 Friston-style variational free energy F is minimized (§ 5.5).

According to our theory of consciousness, consciousness becomes nei-
 ther reduced to a structural property, a functional abstraction, a regime of
 integrated information processing, nor an irreducible primitive. Instead, it
 can be located in a specific regime of organized dissipation, grounded in ther-
 modynamic, metabolic, information-theoretic, thermal, and energetic consid-
 erations. Instead of asking why certain processes – biophysical, functional,
 or computational – give rise to subjective experience at all, we ask about
 the conditions under which certain dissipative systems exhibit the struc-
 tured, integrated, yet flexible dynamics characteristic of conscious states.
 These conscious states, we argue, occupy a constrained region in a joint
 thermodynamic-informational space, governed by free energy minimization
 and parametrized by measurable physiological quantities such as CMRO₂.
 Through linking metabolix flux, the integration of information, and entropy
 within a single formalism, our account strives to render these conscious states
 continuous with the general physics of non-equilibrium systems, while re-
 maining sensitive to the phenomenological distinctions that motivated the
 hard problem in the first instance.

§2 Entropy production in living organisms

A living cell is an example of an open system: it can exchange both energy (heat, radiation, mechanical work, and so on) and matter (nutrients, ions, waste, and so on) with its surroundings.¹ The total entropy change for such an open system can be defined as:

$\frac{dS_{\text{total}}}{dt} = \frac{dS_i}{dt} + \frac{dS_e}{dt}$, where $\frac{dS_{\text{total}}}{dt}$ denotes the total entropy change, $\frac{dS_i}{dt}$ denotes the internal entropy production rate (generated by processes such as friction, diffusion, chemical reactions, and so on) and $\frac{dS_e}{dt}$ denotes the entropy exchange rate with the surroundings (heat transfer, matter entering or leaving the system, and so on)

According to the second law of thermodynamics, the internal entropy production rate $\frac{dS_i}{dt} = \sigma \geq 0$.² The total entropy change for an open system can therefore be generalized under the second law as:

$$\frac{dS_{\text{total}}}{dt} = \sigma + \frac{dS_e}{dt}, \text{ where } \sigma \geq 0$$

If an open system exports enough entropy to its surroundings ($\frac{dS_e}{dt} < 0$), it may have $\frac{dS_{\text{total}}}{dt} < 0$, while still satisfying $\sigma \geq 0$ in terms of its internal entropy production. For many steady-state open systems, $\frac{dS_{\text{total}}}{dt} = 0$. From $0 = \sigma + \frac{dS_e}{dt}$, we can infer that $\frac{dS_e}{dt} = -\sigma$. Living organisms, as an example of open systems, therefore maintain local order by producing entropy internally

¹ Other examples of open systems include a boiling pot without a lid, an engine, and our planet Earth. By contrast, closed systems exchange energy but not matter with their surroundings, whereas isolated systems exchange neither energy nor matter with their surroundings.

² The second law of thermodynamics states that the state of disorder or entropy of a system tends to increase over time (Boltzmann, 1910; Planck, 1914).

(σ) and exporting it to its surroundings ($\frac{dS_e}{dt} = -\sigma$), resulting in an entropy balance. Life requires the following:

(C1 or non-equilibrium steady state) $\sigma + \frac{dS_e}{dt} \approx 0$

(C2 or negative entropy) $\frac{dS_e}{dt} \approx -\sigma < 0$ – from C1 and closely connected to Schrödinger’s (2025) concept of negative entropy and life maintaining its local order by exporting entropy

(C3 or dissipative regime) $\sigma \gg 0$ – life exists in a far-from-thermodynamic-equilibrium, dissipative regime and maintains $\sigma \gg 0$ through a continuous throughput exchange of matter and energy (Prigogine, 1978, 1980; Prigogine and Nicolis, 1985; Prigogine and Stengers, 2018)

Let P denote the metabolic power (measured in watts or W) or the rate at which a body converts chemical energy (food, ATP, and so on) into work and heat. In addition, let T denote the absolute temperature (measured in kelvins or K) or body temperature of an organism. The total entropy rate $\dot{S}$ can be represented in the following manner:

$$\dot{S} \sim \frac{P}{T}$$

In a steady state, $\dot{S}$ – the entropy production/export rate associated with metabolism – is approximately equal to σ : $\dot{S} \approx \sigma$. In biology discussions, $\frac{P}{T}$ becomes a coarse-grained estimate for the more fundamental thermodynamic quantity σ (internal entropy production rate). For a resting human being, $P \approx 80\text{--}120$ W or Js^{-1} and $T \approx 310$ K or 37°C . A back-of-the-envelope estimate for the internal entropy production rate σ of a resting human being will therefore be on the order of:

$$\sigma \approx \dot{S} \sim \frac{100\text{W}}{310\text{K}} \text{ or } 0.3 \text{ WK}^{-1}$$

As a result of higher metabolic activity, P can be raised to 300W.³ This raises $\dot{S}$ and σ toward $\frac{300\text{W}}{310\text{K}}$ or 1 WK⁻¹.

§3 Oxygen consumption in the brain

σ denotes the internal entropy production rate or entropy produced by a living organism per unit volume per unit of time (§ 2). The total or whole-body entropy production rate is therefore the volume integral of σ :

$$\sigma_{\text{w.b.}} = \int_V \sigma dv$$

Whole-body entropy production rate ($\sigma_{\text{w.b.}}$) is commonly approximated from $\dot{V}\text{O}_2$, since oxygen consumption reflects the rate of exergonic redox reactions or catabolic reactions (see also fn 3). At the same time, $\dot{V}\text{O}_2$ is not equivalent to entropy production, since entropy production also depends on other irreversible processes that leave no O₂ footprint: ion pumps leaking across membranes, far-from-equilibrium biosynthetic reactions, and so on. Therefore, the approximation is:

³ $\dot{V}\text{O}_{2\text{ max}}$ (the maximal rate of oxygen consumption attainable during physical exertion) represents the 100% maximal aerobic capacity of a human being. $\dot{V}\text{O}_2$ (measured in mL·kg⁻¹·min⁻¹) can be calculated using the equation:

$$\dot{V}\text{O}_2 = 1.8 \times \frac{\text{work rate}}{\text{body mass}} + 7$$

We can derive work rate (measured in kg·m·min⁻¹) = $(\dot{V}\text{O}_2 - 7) \times \frac{\text{body mass}}{1.8}$. We can then derive the power (measured by W) from work rate by dividing by 6:

$$P = (\dot{V}\text{O}_2 - 7) \times \frac{\text{body mass}}{1.8 \times 6}$$

To sustain $P = 400\text{W}$, you need a $\dot{V}\text{O}_{2\text{ max}}$ of roughly 65–69 (depending on body mass), consistent with elite or professional cyclists. $P = 200\text{W}$ sits in the trained recreational range, whereas $P = 300\text{W}$ sits in the well-trained or competitive range (Barstow and Molé, 1991; Koppo et al., 2004; Sports Medicine et al., 2013).

$\sigma_{\text{w.b.}} \approx (\dot{V}\text{O}_2)_{\text{w.b.}} \cdot \frac{\Delta G_{\text{O}_2}}{T}$, where $\sigma_{\text{w.b.}}$ denotes the whole-body entropy production rate, $(\dot{V}\text{O}_2)_{\text{w.b.}}$ denotes the whole-body oxygen consumption rate as a proxy for aerobic metabolic flux, ΔG_{O_2} denotes the free energy released per mole of O_2 , and T denotes the absolute temperature⁴

Next, suppose that $i \in \{\text{brain, heart, liver, kidneys, skeletal muscle}, \dots\}$ is an index over organs, with each organ or tissue compartment contributing to whole-body oxygen consumption $((\dot{V}\text{O}_2)_{\text{w.b.}})$. Assuming that these organs partition the whole-body $\dot{V}\text{O}_2$ without overlap, the whole-body sum $\sigma_{\text{w.b.}}$ can be reconstructed as:

$$\sigma_{\text{w.b.}} = \sum_i \sigma_i \approx \sum_i (\dot{V}\text{O}_2)_i \cdot \frac{\Delta G_{\text{O}_2}}{T}$$

Fick's (1870) principle allows us to go from whole-body to organ-specific contributions for each organ i :

$(\dot{V}\text{O}_2)_i = \text{CBF}_i \cdot (\text{Ca} - \text{Cv})$, where CBF_i denotes the blood flow to organ i , Ca denotes the arterial oxygen content, and Cv denotes the venous oxygen content leaving organ i . $(\text{Ca} - \text{Cv})$ is also known as the arteriovenous oxygen difference

Through a combination of Fick's principle, the partitioning of the whole body into non-overlapping organ compartments, regional blood flow data, and arteriovenous oxygen differences, resting $\dot{V}\text{O}_2$ can be assigned to the brain (20%), heart (11%), liver (17%), kidneys (6%), skeletal muscle (20%), and other tissues (Rolfe and Brown, 1997). These resting $\dot{V}\text{O}_2$ fractions hold

⁴ $\Delta G_{\text{O}_2} \approx -470\text{kJ}\cdot\text{mol}^{-1}\text{O}_2$ at physiological conditions (Gnaiger and Kemp, 1990).

under resting, steady-state conditions and shift substantially with changes in the physiological state. For instance, skeletal muscle will rise to dominate whole-body $\dot{V}O_2$ during exercise. For the brain, the $(\dot{V}O_2)_i$ can be rewritten as:

$$CMRO_2 = CBF_{\text{brain}} \cdot (C_a - C_v), \text{ where } CMRO_2 \text{ denotes the cerebral metabolic rate of oxygen}$$

§4 Describing different brain states in terms of the normalized ratio

It is standard neurophysiology to approximate the normal, resting value for $CMRO_2$ in an adult as $3.0\text{--}3.5 \text{ mL O}_2 \cdot 100\text{g}^{-1} \cdot \text{min}^{-1}$ (Kety, Schmidt, et al., 1948; Lassen, 1959; Madsen, Holm, et al., 1993). More generally and for any brain state i :

$$\begin{aligned} \text{(Normalization)} \quad CMRO_2^{\text{wake}} &\approx 3.0\text{--}3.5 \text{ mL O}_2 \cdot 100\text{g}^{-1} \cdot \text{min}^{-1} \\ \text{(Normalized ratio)} \quad \mathcal{R} &= \frac{CMRO_2^i}{CMRO_2^{\text{wake}}} \end{aligned}$$

$3.0\text{--}3.5 \text{ mL O}_2 \cdot 100\text{g}^{-1} \cdot \text{min}^{-1}$ can therefore be normalized as the quiet, waking-state baseline ($\mathcal{R} = 1$). During active cognition and vibrotactile stimulation, a regional (non-whole-brain) $CMRO_2$ increase of up to 5% has been observed in the activated somatosensory region: this applies to a small cortical patch and does not reflect a whole-brain change (Fox and Raichle, 1986). Nonetheless and at the global level, whole-brain oxidative metabolism is remarkably consistent across a wide range of mental and motor tasks (Raichle and Gusnard, 2002). Direct Kety–Schmidt measurement during the Wisconsin Card Sorting Test has confirmed that global $CMRO_2$ remains statistically

unchanged (-1%) (Madsen, Hasselbalch, et al., 1995). The normalized ratio for active cognition is therefore $\mathcal{R} \approx 1$ at the whole-brain level, with a regional upper bound of $\mathcal{R} \approx 1.05$ when confined to the locally activated cortex. A seizure is the most metabolically extreme physiological state that a brain can enter: it consists of a sudden surge of electrical activity in the brain. A threefold regional increase in CMRO_2 has been reported during seizure activity (Katsura et al., 1994). This implies a regional upper bound of $\mathcal{R} \approx 3$ relative to the baseline.

During rapid-eye-movement or REM sleep, CMRO_2 has been observed to be practically the same as in the normal (wakeful) state: $\mathcal{R} \approx 1$ (Madsen, Schmidt, et al., 1991). The CMRO_2 associated with the REM sleep state is therefore consistent with the electrophysiological and behavioral activation characteristic of the normal, wakeful state. By contrast, deep slow-wave sleep (non-rapid-eye-movement or NREM stage 3) is associated with a 25% decrease in CMRO_2 relative to the waking state: $\mathcal{R} \approx 0.75$ (Madsen, Schmidt, et al., 1991). Burst suppression – an electroencephalography or EEG pattern where periods of high-voltage electrical activity alternate with periods of no activity – indicates highly depressed cerebral activity and metabolism. This EEG pattern can be found in brain states associated with deep general anaesthesia (especially high-dose propofol), severe hypothermia, and coma. Studies suggest CMRO_2 reductions of approximately 30–70% (Artru, Shapira, and Bowdle, 1992; Siddiqi et al., 2023; Werner et al., 1992). In other words, $\mathcal{R} \approx 0.3\text{--}0.7$, depending on anaesthetic depth, temperature, aetiology, degree of burst suppression, and measurement method.⁵

Studies on the state of hibernation or deep torpor typically track whole-organism oxygen consumption or $(\dot{\text{V}}\text{O}_2)_{\text{w.b.}}$ rather than brain-specific oxygen

⁵ Ordinary surgical anaesthesia is usually titrated to a lighter level, thereby minimizing or avoiding burst suppression.

consumption or CMRO_2 . Nonetheless, as brain metabolism is roughly parallel with whole-body metabolism during deep torpor, we can estimate the normalized cerebral metabolic suppression. Crucially, deep torpor or hibernation produces some of the lowest metabolic rates observed in mammals, short of death: metabolism in mammals can be suppressed by roughly 90–99 % relative to normal (euthermic, wakeful) metabolism (Heldmaier, Ortman, and Elvert, 2004; Wang, 1978, 1989). This implies a normalized ratio of $\mathcal{R} \approx 0.01\text{--}0.1$. By contrast, brain death is defined as the irreversible loss of all functions of the entire brain (Posner et al., 2008, p. 8). Cerebral angiography, the gold standard for assessing cerebral blood flow, confirms brain death by revealing an absence of blood flow to the brain. If $\text{CBF}_{\text{brain}} \rightarrow 0$ and $\text{CMRO}_2 = \text{CBF}_{\text{brain}} \cdot (C_a - C_v)$ by Fick’s principle, then $\text{CMRO}_2 \rightarrow 0$ as a necessary consequence.

Our results concerning the different brain states can be compiled into the following table:

§5 The postulates of consciousness

5.1 The first postulate

We arrive at our first postulate of consciousness:

(First postulate) The intensity of subjective experience $C_{\text{intensity}}$ correlates with thermodynamic distance $\mathcal{D}$ from equilibrium. Formally, $C_{\text{intensity}} \propto \mathcal{R}$.

The crucial move involves recognizing how the normalized cerebral metabolic ratio $\mathcal{R}$, first introduced in § 3, is already a proxy for $\mathcal{D}$ – not

Table 1: An entropy production spectrum for the brain, based on the normalized cerebral metabolic ratio $\mathcal{R} = \text{CMRO}_2^i / \text{CMRO}_2^{\text{wake}}$ across brain states

Brain state	$\mathcal{R}$
Brain death	$\rightarrow 0$
Deep torpor or hibernation	0.01–0.10
Burst suppression (deep anaesthesia, severe hypothermia, coma)	0.30–0.70
Deep slow-wave sleep (NREM stage 3)	≈ 0.75
REM sleep	≈ 1.00
Quiet waking (baseline, $\mathcal{R} \equiv 1$)	1.00
Active cognition (whole-brain)	≈ 1.00
Active cognition (regional upper bound)	≈ 1.05
Seizure (regional upper bound)	≈ 3.00

merely a metabolic index, but a normalized measure of how far the dissipative dynamics of the brain (as tracked by CMRO_2 or oxygen consumption) depart from the thermodynamic ground state of brain death ($\mathcal{R} \rightarrow 0$). Since $\mathcal{R} = \text{CMRO}_2^i / \text{CMRO}_2^{\text{wake}}$ is a linear rescaling of the absolute metabolic rate, we have:

$$\mathcal{R} \sim \frac{\mathcal{D}}{\mathcal{D}_{\text{wake}}}$$

Table 1 already encodes $\mathcal{D}$ in normalized form. Indeed, our table licenses a phenomenological reading: the metabolic ordering along an entropy production spectrum ($\mathcal{R} : 0 \nearrow 1$) coincides exactly with the phenomenological ordering from the total absence of subjective experience through graded unconsciousness to full waking presence. Brain death and, to a lesser extent, deep torpor or hibernation are not merely reduced experiences: we have good

clinical and theoretical reason to treat them as the near- or complete absence of subjective experience ($\mathcal{R} \rightarrow 0$). Burst suppression occupies a contested zone and involves states where there is a lack of clinical certainty concerning phenomenal presence (0.30–0.70). The wide metabolic range simply mirrors this phenomenological indeterminacy.

NREM stage 3 is a philosophically important state. There is strong evidence for reduced but non-zero subjective experience in this state: lingering fragments of mental imagery, sensations, or thoughts during the hypnagogic state, the non-negligible rate of dream recall under careful investigation, and so on. This is duly reflected in $\mathcal{R} \approx 0.75$, which places NREM stage 3 at about three-quarters distance to the waking state. REM and waking states coincide at $\mathcal{R} \approx 1.00$. This is philosophically significant, since REM dreaming, vivid and phenomenally saturated, carries roughly the same thermodynamic signature as the baseline quiet waking state ($\mathcal{R} \equiv 1$). Our phenomenological reading predicts a possible equivalence of experiential intensity between REM dreaming and waking, and this prediction matches phenomenological report. Finally, active cognition barely exceeds the baseline. The first postulate of consciousness does not imply that effortful cognition floods consciousness with intensity $C_{\text{intensity}}$. Indeed, regional hotspots reach only an upper bound of $\mathcal{R} \approx 1.05$.

The upshot of our phenomenological reading is that the brain is an organ that runs almost constantly at its thermodynamic ceiling. The range of $\mathcal{R}$ is neither $[0,10]$ nor even $[0,3]$. Instead, its range is effectively $[0,1]$, with the quiet waking state as a natural maximum. Waking-state consciousness does not occupy some intermediate position on an open-ended scale of experiential intensity $C_{\text{intensity}}$. Instead, it sits at the normal physiological limit of what neural tissue, bound by oxygen delivery, thermal dissipation, and other related neurophysiological constraints, can maintain indefinitely. The

metabolic rate can increase 100-fold from rest to exercise in active skeletal muscle, and higher rates of increase in short bursts have been observed in fast-twitch fibres (Hargreaves and Spriet, 2020). Unlike skeletal muscle, however, the brain appears to operate almost permanently near its sustainable maximum.

That seizure states reach a regional upper bound of $\mathcal{R} \approx 3$ is consistent with our account: they are pathological precisely because they briefly exceed what the system can sustain. Furthermore, our first postulate suggests that these seizure states should be accompanied by correspondingly elevated experiential intensity $C_{\text{intensity}} \gg 0$. This is supported by evidence that focal seizures are often described as among the most intense experiences of a person’s life (increased self-awareness, déjà vu, jamais vu, sudden floods of emotion, sensory distortions and hyperreal perception, and temporal distortion) (Picard, 2023; Picard and Craig, 2009). The canonical literary witness of seizure states is the Russian novelist Fyodor Dostoevsky, who is believed to have had temporal lobe epilepsy (Rayport, Rayport, and Schell, 2011). Dostoevsky provided among the first detailed phenomenological descriptions of ecstatic auras or feelings of extraordinary harmony, beauty, and plenitude that preceded his seizures. Dostoevsky used his seizure experiences for the character of Prince Myshkin in his novel *The Idiot*, and one of its passages in particular has become a canonical literary description of a pre-seizure state:⁶

He remembered that during his epileptic fits, or rather immediately preceding them, he had always experienced a moment or two when his whole heart, and mind, and body seemed to wake up to vigour and light; when he became filled with joy and hope,

⁶ This passage is not only unusually detailed and precise: it is phenomenologically consistent with our account of experiential intensity.

and all his anxieties seemed to be swept away for ever; these moments were but presentiments, as it were, of the one final second (it was never more than a second) in which the fit came upon him. That second, of course, was inexpressible. When his attack was over, and the prince reflected on his symptoms, he used to say to himself: ‘These moments, short as they are, when I feel such extreme consciousness of myself, and consequently more of life than at other times, are due only to the disease – to the sudden rupture of normal conditions. Therefore they are not really a higher kind of life, but a lower.’ This reasoning, however, seemed to end in a paradox . . . (Dostoevsky, 1900, p. 327)

5.2 The Arrhenius-style corollary of the first postulate

As a corollary of the first postulate, the Arrhenius equation governs the rate of biochemical reactions underlying brain-specific metabolism:

$k = Ae^{-\frac{E_a}{RT}}$, where k denotes the reaction rate constant, E_a denotes the activation energy, A denotes the Arrhenius factor, R denotes the universal gas constant, and T denotes the absolute temperature (Arrhenius, 1889a,b)

The relevance here is that CMRO_2 itself is temperature-dependent: brain metabolism decreases as temperature decreases. For each 1°C decrease in body temperature, CMRO_2 decreases by about 7% (Hemmings and Hopkins, 2006, p. 248).⁷ This implies that $\mathcal{R}$ does not merely function as a

⁷ The metabolic temperature Q_{10} is defined as the ratio between CMRO_2 at a given temperature T and CMRO_2 at temperature $T - 10^\circ\text{C}$. The cerebral $Q_{10} \approx 2.0 - 3.0$ between 37°C and 27°C . Below 27°C , the cerebral Q_{10} increases to 4.5.

normalized ratio for oxygen consumption and cerebral metabolism: it also implicitly encodes an Arrhenius-style thermal constraint on consciousness itself.

(Arrhenius-style corollary of the first postulate) Since neural biochemical reactions obey Arrhenius kinetics, $\mathcal{R}$ implicitly encodes a temperature dependence into the processes underlying consciousness itself through the Q_{10} effect: CMRO_2 falls by roughly 7% per $^{\circ}\text{C}$ of cooling over the moderate hypothermic range.

Through this Arrhenius-style corollary of the first postulate, altered states associated with deep hibernation in heterothermic mammals (typically $0.0\text{--}10.0^{\circ}\text{C}$), severe hypothermia ($< 28.0^{\circ}\text{C}$), fever ($\geq 38.0^{\circ}\text{C}$), and hyperpyrexia ($\geq 41.0^{\circ}\text{C}$) can now be interpreted as modulations of $C_{\text{intensity}}$, mediated in part by temperature-dependent changes in reaction kinetics and cerebral metabolism. By contrast, the optimal temperature range for consciousness is relatively narrow at $36.5\text{--}37.5^{\circ}\text{C}$.

5.3 The second postulate

At the same time, CMRO_2 and its associated, normalized cerebral metabolic ratio $\mathcal{R}$ alone do not determine consciousness, since psychedelics and individuals in normal waking states have a similar metabolism but different phenomenologies. Indeed, the hallucinogenic compound psilocybin – found in hundreds of species of mushrooms – has been found to decrease global cerebral blood flow or $\text{CBF}_{\text{brain}}$, even as reported subjective experience remained vivid and phenomenologically distinct (Carhart-Harris, Erritzoe, et al., 2012). For convenience, we may term this the psychedelic paradox. The psychedelic paradox cannot be resolved by the first postulate (§ 5.1) alone.

Despite near-normal CMRO_2 ($\mathcal{R} \approx 1$), psychedelic consciousness shifts toward high informational entropy (H) at the expense of integration (Φ): the resulting subjective experience, though intense (good metabolism), is fragmented (poor Φ – H balance). The intense though disintegrated experience that psychedelics typically report feeling suggest that metabolism is necessary though insufficient for explaining consciousness. The first postulate ensures that we need energy to be conscious at all: the intensity of subjective experience scales with metabolism. Another postulate, however, is required to resolve the psychedelic paradox and account for the richness, quality, coherence, and organization of that experience.

This leads us to our second postulate of consciousness:

(Second postulate) The quality and coherence of subjective experience depends on a balance between the integration (Φ) and entropy (H) of information. Formally, $C_{\text{quality}} = f(x)$, where x represents the Φ – H balance.

The second postulate is consistent with the first: rich conscious states typically require more metabolic energy than simple ones, and you cannot maintain high Φ or high H without a sufficient level of oxygen metabolism (CMRO_2) to support the underlying neural dynamics. Deep anaesthesia is typically associated with a low Φ , low H , and minimal or absent conscious experience (Leung et al., 2021; Schartner, Seth, et al., 2015). The normal, waking state is typically associated with high Φ , moderate H (compared to low H in sleep or anaesthesia and high H in psychedelia), and a stable, integrated consciousness (Mediano et al., 2021; Tononi and Edelman, 1998).

Psychedelic states are typically associated with moderate Φ , very high H , and a vivid yet fragmented subjective experience (Carhart-Harris, Leech,

et al., 2014; Schartner, Carhart-Harris, et al., 2017). Flow states are typically associated with very high Φ (compared to high Φ the normal, waking state), moderate H , and a unified, absorbed subjective experience (Huskey, Wilcox, and Weber, 2018; Shehata et al., 2021). Focused, meditative states are typically associated with very high Φ , moderate to low H , and stable, focused attention (Lutz et al., 2004; Tang, Hölzel, and Posner, 2015). States of delirium or fever are typically associated with low Φ , high H , and the experience of confusion and disorientation (Sanders, 2011; Van Montfort et al., 2019). We have a neat account of why delirium or fever, though sitting at $\mathcal{R} \approx 1$, still produces a disturbed consciousness: the temperature shifts the Arrhenius landscape (§ 5.2), increasing the cerebral metabolic rate but destabilizing the Φ – H balance, thereby feeding directly into our second postulate.

All things considered, the brain is a dissipative organ, operating far from thermodynamic equilibrium. Its conscious dynamics require ion pumping, synaptic transmission, the maintenance of attractors, and so on. All these processes consume oxygen, making CMRO_2 – itself already a useful proxy for metabolic throughput, as reflected in $(\dot{V}\text{O}_2)_{\text{w.b.}}$, and entropy production σ – a further proxy for the capacity of a system to explore and sustain trajectories within a consciousness state space. A sufficiently rich and coherent level of consciousness requires a sufficient level of cerebral metabolic throughput (fuel; $\text{CMRO}_2 \approx 3.0\text{--}3.5 \text{ mL O}_2 \cdot 100\text{g}^{-1} \cdot \text{min}^{-1}$ and $\mathcal{R} \approx 1$), operating at an optimal Φ – H balance (organization). These observations yield a consciousness state space that is illustrated by the table below:

Table 2: A consciousness state space populated by different brain states, relative to a balance between Φ and H and the metabolic baseline $\mathcal{R} \equiv 1$. The Φ and H characterizations are qualitative and ordinal rather than quantitative in nature.

Brain state	Φ	H	$\mathcal{R}$	Phenomenology
Deep anaesthesia	Low	Low	0.30–0.70	Minimal or absent experience
Quiet waking	High	Moderate	1.00	Stable, integrated experience
Psychedelic state	Moderate	Very high	≈ 1.00	Vivid yet fragmented experience
Flow state	Very high	Moderate	≈ 1.00	Unified, absorbed experience
Meditative state	Very high	Moderate to low	≈ 1.00	Stable, focused attention
State of delirium or fever	Low	High	≈ 1.00	Confusion, disorientation

5.4 The third postulate

A further postulate allows us to formalize the probability distribution of consciousness states over the Φ – H state space:

(Third postulate) Let the normalized ratio $\mathcal{R}$ be modelled as a Beta-distributed random variable $\mathcal{R} \sim \text{Beta}(\alpha, \beta)$, where α and β can encode Φ – H balance. $\{\text{High } \alpha, \text{low } \beta\}$ corresponds to states

of high Φ and low H (flow and meditative states), whereas $\{\text{low } \alpha, \text{high } \beta\}$ corresponds to states of low Φ and high H (delirium and psychedelia).

The Beta distribution $\text{Beta}(\alpha, \beta)$ is defined on $[0, 1]$ and serves as a natural choice for modelling $\mathcal{R}$ as a random variable, since $\mathcal{R} \in [0, 1]$ by construction. The mean $\mu = \frac{\alpha}{\alpha+\beta}$ recovers the expected metabolic position $\mathcal{R}$ for each state, while the variance $\sigma^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ encodes the phenomenological stability of that state. Flow and meditative states have a high α , low β , a mean μ that is biased toward the Φ -dominated side, and a low variance (high concentration of $\mathcal{R}$ around μ). These states can be interpreted in terms of their expected metabolic position (consistently in the high- Φ , low- H regime) and phenomenological stability ($\mathcal{R}$ is locked in and fluctuations are small). By contrast, delirium and psychedelia have a low α , high β , a mean μ that is biased toward the H -dominated side, and a high variance (broad distribution of $\mathcal{R}$). These states can be interpreted in terms of their expected metabolic position (consistently in the low- Φ , high- H regime) and phenomenological instability ($\mathcal{R}$ is volatile and each system makes large excursions across the state space).

5.5 The fourth postulate

A fourth postulate allows us to connect thermodynamic entropy S , governing the physical dissipation costs encoded in $\mathcal{R}$ (§ 5.1, § 5.2, § 5.4), informational entropy H , governing the diversity of brain states (§ 5.3), and Friston-style variational free energy F . According to the active inference framework, the brain, as an active inference machine, minimizes variational free energy F (Adams, Shipp, and Friston, 2013; Carhart-Harris and Friston, 2010; Clark, 2013; Friston, 2005, 2009; Friston and Stephan, 2007; Hohwy, 2013). A

brain that minimizes F simultaneously minimizes informational surprise (reducing informational entropy or H to an optimal level) and regulates its thermodynamic costs (maintaining $\mathcal{R} \approx 1$). Through the fourth postulate, thermodynamic dissipation (tracked by $\mathcal{R}$) and informational free energy minimization (tracked by Φ , H , and the Φ – H balance) are formally related in the following manner:

(Fourth postulate) Conscious states occupy regions in the Φ – H state space in which F is minimized.⁸ High Φ ensures that the internal model is well-integrated and coherent, while moderate H ensures that this model remains sufficiently flexible to accommodate new sensory data. The metabolic cost of this free energy minimization is tracked by $\mathcal{R}$, connecting thermodynamic entropy and informational entropy within a single dissipative framework.

§6 Conclusion

In conclusion, our account proposes that consciousness is best understood as a thermodynamically grounded, informationally structured phenomenon that emerges in certain dissipative systems that are capable of maintaining local order far from equilibrium. By linking the intensity of subjective experience $C_{\text{intensity}}$ to thermodynamic distance $\mathcal{D}$ (§ 5.1) and the quality and coherence of this experience to the balance between informational integration Φ and informational entropy H (§ 5.3), our framework unifies metabolic, information-theoretic, and phenomenological dimensions within a single for-

⁸ A brain that minimizes F implicitly bounds information surprise and constrains H to levels compatible with its own structural and environmental constraints.

mal scheme. In addition, the Beta-distributed parametrization of $\mathcal{R}$ embeds conscious states within a continuous and quantifiable state space governed by Friston-style variational free energy minimization (§§ 5.4–5.5).

We do not claim to have eliminated the full conceptual force of the hard problem of consciousness. We do however believe that our account forces a reframing of the scope of this problem. In particular, the explanatory gap is no longer to be naively construed as a divide between the biophysical and the experiential, but rather as a question concerning why certain regions of non-equilibrium, free-energy-minimizing dynamics correspond to structured subjective experiences. In this sense, consciousness is neither reduced to a structural property, a functional abstraction, a regime of integrated information processing, nor an irreducible primitive. Instead, it emerges as a characteristic regime of dissipative organization, grounded in thermodynamic, metabolic, information-theoretic, thermal, and energetic constraints. What remains to be explained, then, is not some metaphysical mystery but a problem of regime specificity: the task of characterizing, with increasing care and precision, the boundary conditions under which matter, in sustaining itself far from equilibrium, gives rise to an inner life.

References

- Adams, Rick A., Stewart Shipp, and Karl Friston (2013). “Predictions not commands: active inference in the motor system”. In: *Brain Structure & Function* 218, pp. 611–643.
- Albantakis, Larissa et al. (2023). “Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms”. In: *PLoS Computational Biology* 19.10, e1011465.

- Arrhenius, Svante (1889a). In: *Zeitschrift für Physikalische Chemie* 4U.1, pp. 96–116. DOI: doi:10.1515/zpch-1889-0408.
- (1889b). In: *Zeitschrift für Physikalische Chemie* 4U.1, pp. 226–248. DOI: doi:10.1515/zpch-1889-0416.
- Artru, Alan A., Yoram Shapira, and T. Andrew Bowdle (1992). “Electroencephalogram, cerebral metabolic, & vascular responses to propofol anesthesia in dogs”. In: *Journal of Neurosurgical Anesthesiology* 4.2, pp. 99–109.
- Barstow, Thomas J. and Paul A. Molé (1991). “Linear & nonlinear characteristics of oxygen uptake kinetics during heavy exercise”. In: *Journal of Applied Physiology* 71.6, pp. 2099–2106.
- Block, Ned (1995). “On a confusion about a function of consciousness”. In: *Brain & Behavioral Sciences* 18.2, pp. 227–247. DOI: 10.1017/s0140525x00038188.
- Boltzmann, Ludwig (1910). *Vorlesungen über gastheorie*. Vol. 1. J. A. Barth (A. Meiner).
- Carhart-Harris, Robin and Karl Friston (2010). “The default-mode, ego-functions & free-energy: A neurobiological account of Freudian ideas”. In: *Brain* 133.4, pp. 1265–1283.
- Carhart-Harris, Robin L., David Erritzoe, et al. (2012). “Neural correlates of the psychedelic state as determined by fMRI studies with psilocybin”. In: *Proceedings of the National Academy of Sciences* 109.6, pp. 2138–2143.
- Carhart-Harris, Robin Lester, Robert Leech, et al. (2014). “The entropic brain: A theory of conscious states informed by neuroimaging research with psychedelic drugs”. In: *Frontiers in Human Neuroscience* 8, p. 55875.
- Chalmers, David J. (1995). “Facing up to the problem of consciousness”. In: *Journal of Consciousness Studies* 2.3, pp. 200–219.
- (1996). *The Conscious Mind: In Search of a Fundamental Theory*. New York: Oxford University Press.
- Churchland, Paul M. (2013). *Matter & Consciousness*. MIT Press.

- Clark, Andy (2013). “Whatever next? Predictive brains, situated agents, & the future of cognitive science”. In: *Behavioral & Brain Sciences* 36.3, pp. 181–204.
- Crick, Francis, Christof Koch, et al. (1990). “Towards a neurobiological theory of consciousness”. In: *Seminars in the Neurosciences*. Vol. 2. 263–275. London, p. 203.
- Dostoevsky, Fyodor (1900). *The Idiot*. Public Domain Mark 1.0 / Internet Archive.
- Fick, Adolf (1870). “Ueber die Messung des Blutquantum in den Herzventrikeln.” In: *Sitzungsberichte der Physikalisch-Medizinischen Gesellschaft zu Würzburg*, pp. 16–17.
- Fox, Peter T. and Marcus E. Raichle (1986). “Focal physiological uncoupling of cerebral blood flow & oxidative metabolism during somatosensory stimulation in human subjects.” In: *Proceedings of the National Academy of Sciences* 83.4, pp. 1140–1144.
- Friston, Karl (2005). “A theory of cortical responses”. In: *Philosophical Transactions of the Royal Society B: Biological Sciences* 360.1456, pp. 815–836.
- (2009). “The free-energy principle: A rough guide to the brain?” In: *Trends in Cognitive Sciences* 13.7, pp. 293–301.
- Friston, Karl and Klaas E Stephan (2007). “Free-energy & the brain”. In: *Synthese* 159.3, pp. 417–458.
- Gnaiger, Erich and Richard B. Kemp (1990). “Anaerobic metabolism in aerobic mammalian cells: Information from the ratio of calorimetric heat flux & respirometric oxygen flux”. In: *Biochimica et Biophysica Acta - Bioenergetics* 1016.3, pp. 328–332.
- Hargreaves, Mark and Lawrence L. Spriet (2020). “Skeletal muscle energy metabolism during exercise”. In: *Nature Metabolism* 2.9, pp. 817–828.

- Heldmaier, Gerhard, Sylvia Ortmann, and Ralf Elvert (2004). “Natural hypometabolism during hibernation & daily torpor in mammals”. In: *Respiratory Physiology & Neurobiology* 141.3, pp. 317–329.
- Hemmings, Hugh C. and Philip M. Hopkins, eds. (2006). *Foundations of Anesthesia: Basic Sciences for Clinical Practice*. Philadelphia: Mosby Elsevier.
- Hohwy, Jakob (2013). *The Predictive Mind*. Oxford University Press.
- Huskey, Richard, Shelby Wilcox, and René Weber (2018). “Network neuroscience reveals distinct neuromarkers of flow during media use”. In: *Journal of Communication* 68.5, pp. 872–895.
- Katsura, Ken-ichiro et al. (1994). “Functional, metabolic, & circulatory changes associated with seizure activity in the postischemic brain”. In: *Journal of Neurochemistry* 62.4, pp. 1511–1515.
- Kety, Seymour S., Carl F. Schmidt, et al. (1948). “The nitrous oxide method for the quantitative determination of cerebral blood flow in man: Theory, procedure & normal values”. In: *The Journal of Clinical Investigation* 27.4, pp. 476–483.
- Koppo, Katrien et al. (2004). “Overshoot in VO₂ following the onset of moderate-intensity cycle exercise in trained cyclists”. In: *European Journal of Applied Physiology* 93.3, pp. 366–373.
- Lassen, Niels A. (1959). “Cerebral blood flow & oxygen consumption in man”. In: *Physiological Reviews* 39.2, pp. 183–238.
- Leung, Angus et al. (2021). “Integrated information structure collapses with anesthetic loss of conscious arousal in *Drosophila melanogaster*”. In: *PLoS Computational Biology* 17.2, e1008722.
- Lutz, Antoine et al. (2004). “Long-term meditators self-induce high-amplitude gamma synchrony during mental practice”. In: *Proceedings of the National Academy of Sciences* 101.46, pp. 16369–16373.

- Madsen, Peter Lund, Steen G. Hasselbalch, et al. (1995). “Persistent resetting of the cerebral oxygen/glucose uptake ratio by brain activation: Evidence obtained with the Kety—Schmidt technique”. In: *Journal of Cerebral Blood Flow & Metabolism* 15.3, pp. 485–491.
- Madsen, Peter Lund, Søren Holm, et al. (1993). “Average blood flow & oxygen uptake in the human brain during resting wakefulness: A critical appraisal of the Kety—Schmidt technique”. In: *Journal of Cerebral Blood Flow & Metabolism* 13.4, pp. 646–655.
- Madsen, Peter Lund, J. F. Schmidt, et al. (1991). “Cerebral O₂ metabolism & cerebral blood flow in humans during deep & rapid-eye-movement sleep”. In: *Journal of Applied Physiology* 70.6, pp. 2597–2601.
- Mediano, Pedro. et al. (2021). “Fluctuations in neural complexity during wakefulness relate to conscious level & cognition”. In: *BioRxiv*, pp. 2021–09.
- Picard, Fabienne (2023). “Ecstatic or mystical experience through epilepsy”. In: *Journal of Cognitive Neuroscience* 35.9, pp. 1372–1381.
- Picard, Fabienne and A. D. Craig (2009). “Ecstatic epileptic seizures: A potential window on the neural basis for human self-awareness”. In: *Epilepsy & Behavior* 16.3, pp. 539–546.
- Planck, Max (1914). *The Theory of Heat Radiation*. Trans. by M. Masius. P. Blakiston’s Son & Co.
- Posner, Jerome B. et al. (2008). *Plum & Posner’s Diagnosis of Stupor & Coma*. 4th ed. New York: Oxford University Press.
- Prigogine, Ilya (1978). “Time, structure, & fluctuations”. In: *Science* 201.4358, pp. 777–785.
- (1980). *From Being to Becoming: Time & Complexity in the Physical Sciences*. San Francisco: W. H. Freeman & Company.
- Prigogine, Ilya and Grégoire Nicolis (1985). “Self-Organisation in Nonequilibrium Systems: Towards A Dynamics of Complexity”. In: *Bifurcation*

- Analysis: Principles, Applications & Synthesis*. Ed. by M. Hazewinkel, R. Jurkovich, and J. H. P. Paelinck. Dordrecht: Springer Netherlands, pp. 3–12.
- Prigogine, Ilya and Isabelle Stengers (2018). *Order Out of Chaos: Man's New Dialogue with Nature*. Verso Books.
- Putnam, Hilary (1967). "Psychological Predicates". In: *Art, Mind, & Religion*. Ed. by William H. Capitan and Daniel Davy Merrill. University of Pittsburgh Press, pp. 37–48.
- Raichle, Marcus E. and Debra A. Gusnard (2002). "Appraising the brain's energy budget". In: *Proceedings of the National Academy of Sciences* 99.16, pp. 10237–10239.
- Rayport, Shirley M. Ferguson, Mark Rayport, and Carolyn A. Schell (2011). "Dostoevsky's epilepsy: A new approach to retrospective diagnosis". In: *Epilepsy & Behavior* 22.3, pp. 557–570.
- Rolfe, D. F. and Guy C. Brown (1997). "Cellular energy utilization & molecular origin of standard metabolic rate in mammals". In: *Physiological Reviews* 77.3, pp. 731–758.
- Sanders, Robert D. (2011). "Hypothesis for the pathophysiology of delirium: Role of baseline brain network connectivity & changes in inhibitory tone". In: *Medical Hypotheses* 77.1, pp. 140–143.
- Schartner, Michael, Anil Seth, et al. (2015). "Complexity of multi-dimensional spontaneous EEG decreases during propofol induced general anaesthesia". In: *PloS One* 10.8, e0133532.
- Schartner, Michael M., Robin L. Carhart-Harris, et al. (2017). "Increased spontaneous MEG signal diversity for psychoactive doses of ketamine, LSD & psilocybin". In: *Scientific Reports* 7.1, p. 46421.
- Schrödinger, Erwin (2025). *What is Life? The Physical Aspect of the Living Cell*. Rare Treasure Editions.

- Searle, John (2017). “Biological Naturalism”. In: *The Blackwell Companion to Consciousness*, pp. 327–336.
- Shehata, Mohammad et al. (2021). “Team flow is a unique brain state associated with enhanced information integration & interbrain synchrony”. In: *eneuro* 8.5.
- Siddiqi, A. Zohaib et al. (2023). “The effect of burst suppression on cerebral blood flow & autoregulation: A scoping review of the human & animal literature”. In: *Frontiers in Physiology* 14, p. 1204874.
- Sports Medicine, American College of et al. (2013). *ACSM’s Guidelines for Exercise Testing & Prescription*. Lippincott williams & wilkins.
- Tang, Yi-Yuan, Britta K. Hölzel, and Michael I. Posner (2015). “The neuroscience of mindfulness meditation”. In: *Nature Reviews Neuroscience* 16.4, pp. 213–225.
- Tononi, Giulio, Melanie Boly, et al. (2016). “Integrated information theory: From consciousness to its physical substrate”. In: *Nature Reviews Neuroscience* 17.7, pp. 450–461.
- Tononi, Giulio and Gerald M. Edelman (1998). “Consciousness & complexity”. In: *Science* 282.5395, pp. 1846–1851.
- Turing, Alan (1950). “Computing Machinery & Intelligence”. In: *Mind* 59.236, pp. 433–60. DOI: 10.1093/mind/lix.236.433.
- Van Montfort, S. J. T. et al. (2019). “Brain network disintegration as a final common pathway for delirium: A systematic review & qualitative meta-analysis”. In: *NeuroImage: Clinical* 23, p. 101809.
- Wang, L. C. H. (1978). “Energetic& field aspects of mammalian torpor: The Richardson’s ground squirrel”. In: *Strategies in Cold: Natural Torpidity & Thermogenesis*.
- (1989). “Ecological, physiological, & biochemical aspects of torpor in mammals & birds”. In: *Animal Adaptation to Cold*. Springer, pp. 361–401.

Werner, Christian et al. (1992). “The effects of propofol on cerebral blood flow in correlation to cerebral blood flow velocity in dogs”. In: *Journal of Neurosurgical Anesthesiology* 4.1, pp. 41–46.